Jacob Foster

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Education

McMaster University, Hamilton, ON **Sep. 2022 - May 2027**

Bachelor of Mechatronics Engineering - Co-op *cGPA - 3.86/4.00*

- **Relevant Coursework:** AI and ML Principles, Robotics, RTOS, Operating Systems, Advanced Control Systems, Embedded System Design, Analog and Digital Circuits, Software Development and Design, FPGA Design.

Experience

Advanced Micro Devices (AMD) **May 2025 - August 2026**

Video Decoding Software Engineer Intern *Markham, ON*

- Diagnosed and resolved defects in Windows GPU/APU driver software and implemented features across the full driver stack, **from kernel-level to user-level abstractions** using **C/C++.**
- Engineered software for **compressing and decoding** MJPEG video streams via **DCT-based decoding**, enabling hardware acceleration for next-generation Windows GPU, APU, and console drivers across D3D11/12 decode paths.
- Built a bitstream parser that walks JPEG marker structure, detects chroma subsampling, and reformats frames into DXVA-compliant picture parameters and Huffman/quantization tables; enabling the team to validate hardware decoding for product bring-up.
- Designed and built a Python web app that automates the end-to-end firmware submission pipeline, unifying six previously manual stages and reducing submission time by **2.3x**. The tool orchestrates remote **Linux** test machines over **SSH**, drives **Perforce** and **Git** from a single interface, auto-generates and cross-links pull requests across three downstream repositories, and integrates an **AI coding agent** to resolve merge conflicts. Deployed to a team of **30+ developers**.
- Resolved a scheduler defect where all video decode jobs were incorrectly dispatched to the encoding engine; implemented **codec-aware engine routing** to direct workloads to the dedicated decode engine, with AV1 exempt and single-engine APU configurations handled separately, leveraging **GPU decode affinity** to defer engine assignment until hardware capabilities were known at runtime.

L3Harris Technologies - Wescam Division **May 2024 - August 2024**

Manufacturing Engineering Intern *Waterdown, ON*

- Developed a software/firmware comparison web tool to capture current and previous build components of a WESCAM imaging device and automatically determine build status for repairs, **decreasing effective labour time from 20 mins to 1 min**. Built using **C#, .NET Framework, HTML, CSS.**

Projects

ReStride: Wearable Gait Monitor | *Rust, Flutter, Dart, ESP32, BLE, PCB Design* **September 2026**

- Designed a smart insole that measures weight distribution and gait using a **resistive (FSR) sensor array**, validating sensor placement at the heel, midfoot, and forefoot against real walking waveforms.
- Developed **Rust** firmware on an **ESP32** that samples pressure data and streams it over **BLE** from two pods simultaneously with on-device **data buffering**, aloowing for up to **2 days** of gait data between BLE syncs.
- Built a **Flutter/Dart** app and backend that receives both BLE streams, buffers data through connectivity loss, and serves a clinician dashboard with a live pressure heat map, center-of-pressure trajectory, step count, and left/right load symmetry.

ShotTracker | *Python, OpenCV, PyTorch, YOLO, FFmpeg* **October 2026**

- Built a computer-vision pipeline that turns **single-camera footage into per-shot data**: YOLO ball detection, **Kalman-filter tracking** that predicts trajectory through frames where the ball is obstructed , and pose estimation to detect strikes and assign left or right foot.
- Mapped each shot onto the goal plane with a **homography** from the four goal corners and regulation goal dimensions, and rendered a **stats-overlay video with OpenCV** showing the pose skeleton, ball tracer, numbered goal markers, placement map, shot origin and pass-to-shot time, with misses highlighted in red.
- Trained a **placement-based xGOT model** on StatsBomb open data, evaluated against a location-only baseline.

Technical Skills

Computing: Python, C, C++, C#, Rust, Verilog, ffmpeg, Microsoft D3D, Make, HTML, CSS, Javascript, React.js
Developer Tools: Git, GitHub, Claude Code, Cursor, VS Code, Visual Studio, Perforce, Linux, WinDBG, .NET
Technologies: SOLIDWORKS, Autodesk Inventor, Jira, Microsoft Office
Manufacturing: Soldering, 3D printing, Drill press, Hand and power tools